

CCNA Project

Building network infrastructure for an organization

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Basic idea of the project

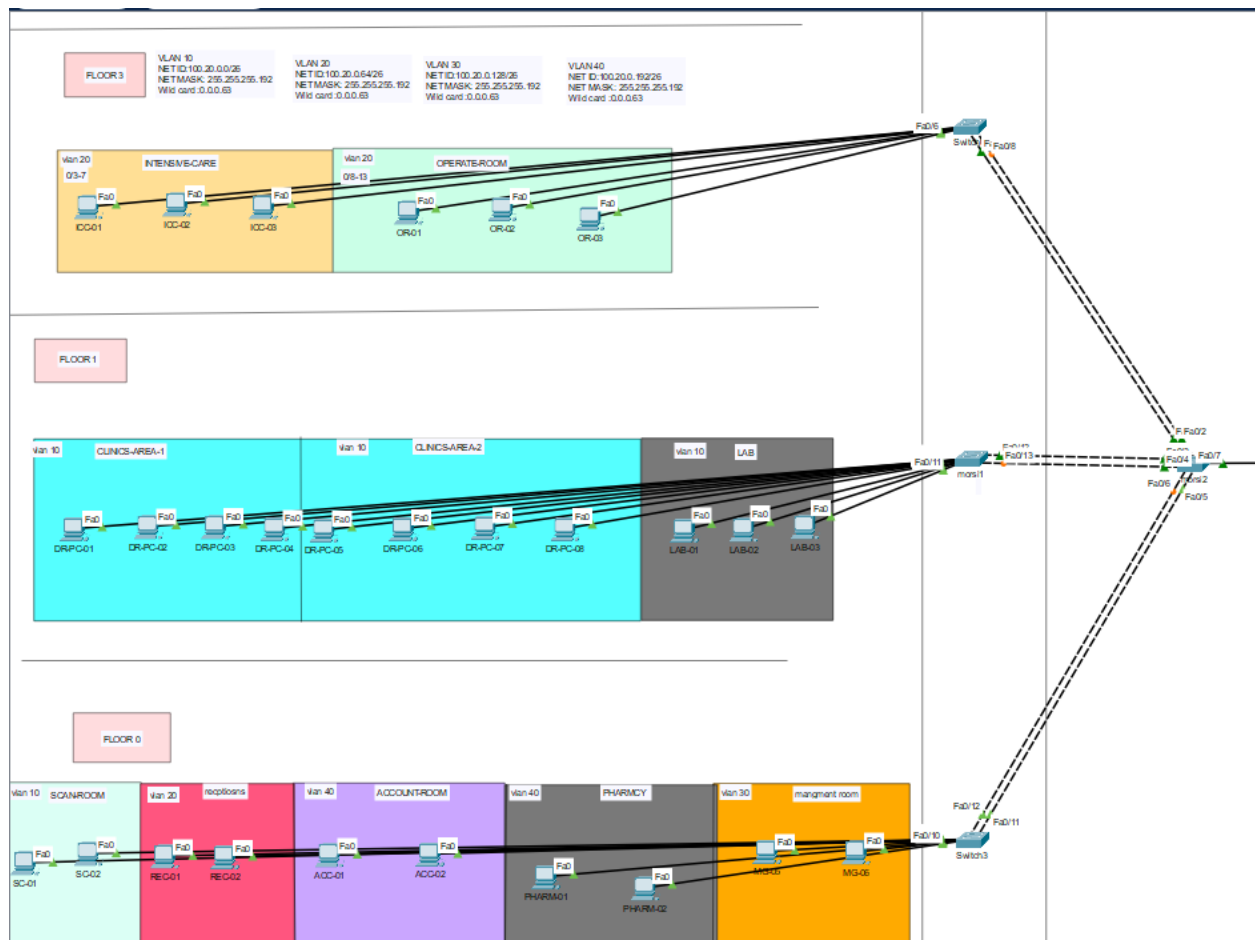
(Hospital Branch Interconnectivity)

This project successfully engineered a unified, high-availability (HA) network connecting two hospital branches to a shared Data Center, which hosts essential services including AAA, DHCP, and FTP. The architecture utilized VLSM for optimal IP address allocation across various segmented VLANs, with Inter-VLAN Routing configured to manage necessary traffic exchange. To centralize and secure all external access, the OSPF dynamic routing protocol was strategically deployed, making the Branch 1 Router the default gateway for all Data Center and Internet-bound traffic originating from Branch 2. Critical network resilience was ensured through two key HA mechanisms: HSRP (Hot Standby Router Protocol) was implemented to provide transparent first-hop redundancy between core devices, and EtherChannel was used to bundle physical links, creating both high-bandwidth capacity and automatic link-level redundancy, thereby guaranteeing continuous, uninterrupted service for the entire hospital system.

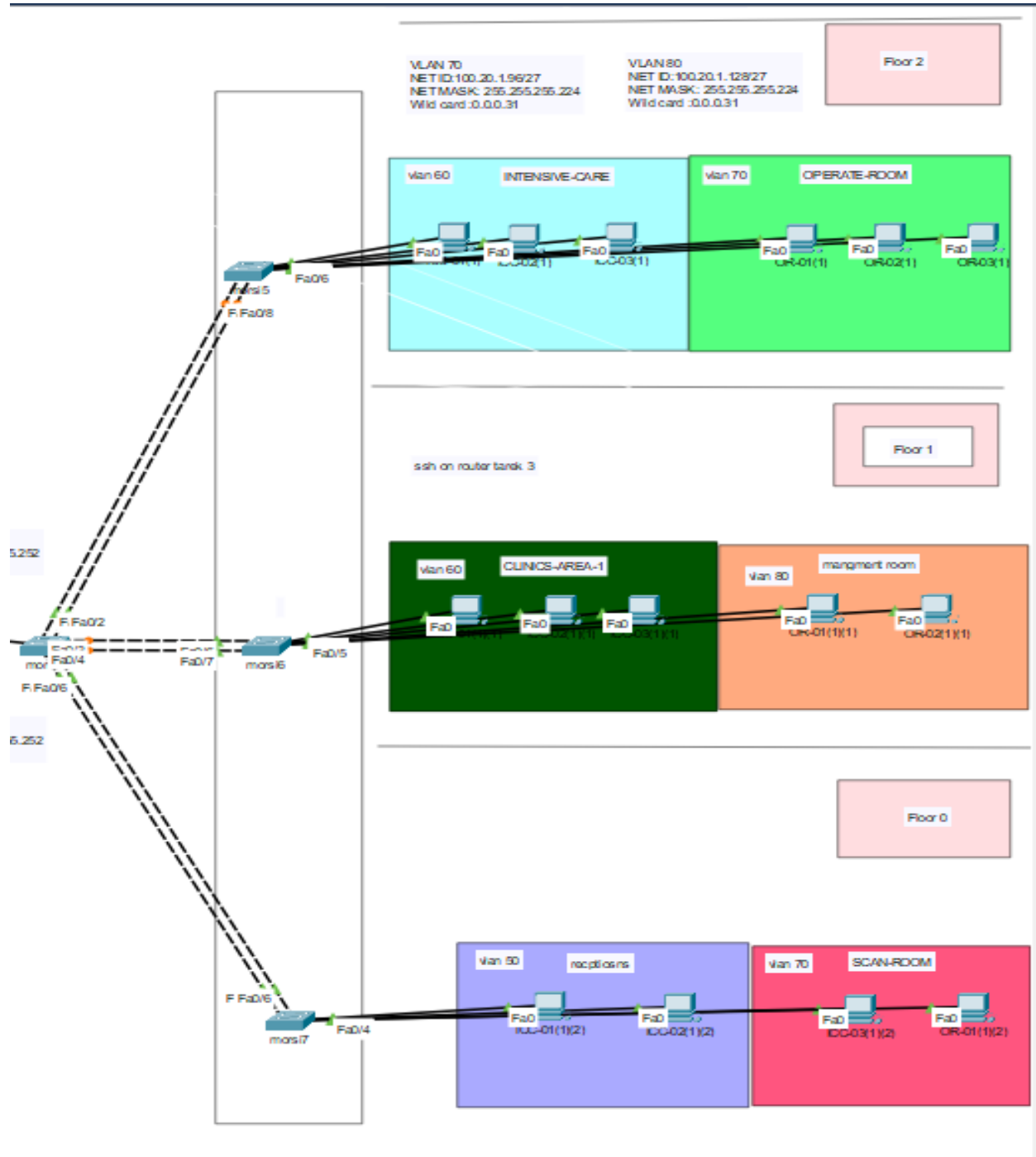
Our Design

We worked on a hospital design where it has two branches each branch contains 3 floors and we created 4 VLANs for each branch and each floor in the branch is connected to a switch and all those floor switches connected to a main switch which goes to the routing part where routers found that connect between two branches and goes on the internet

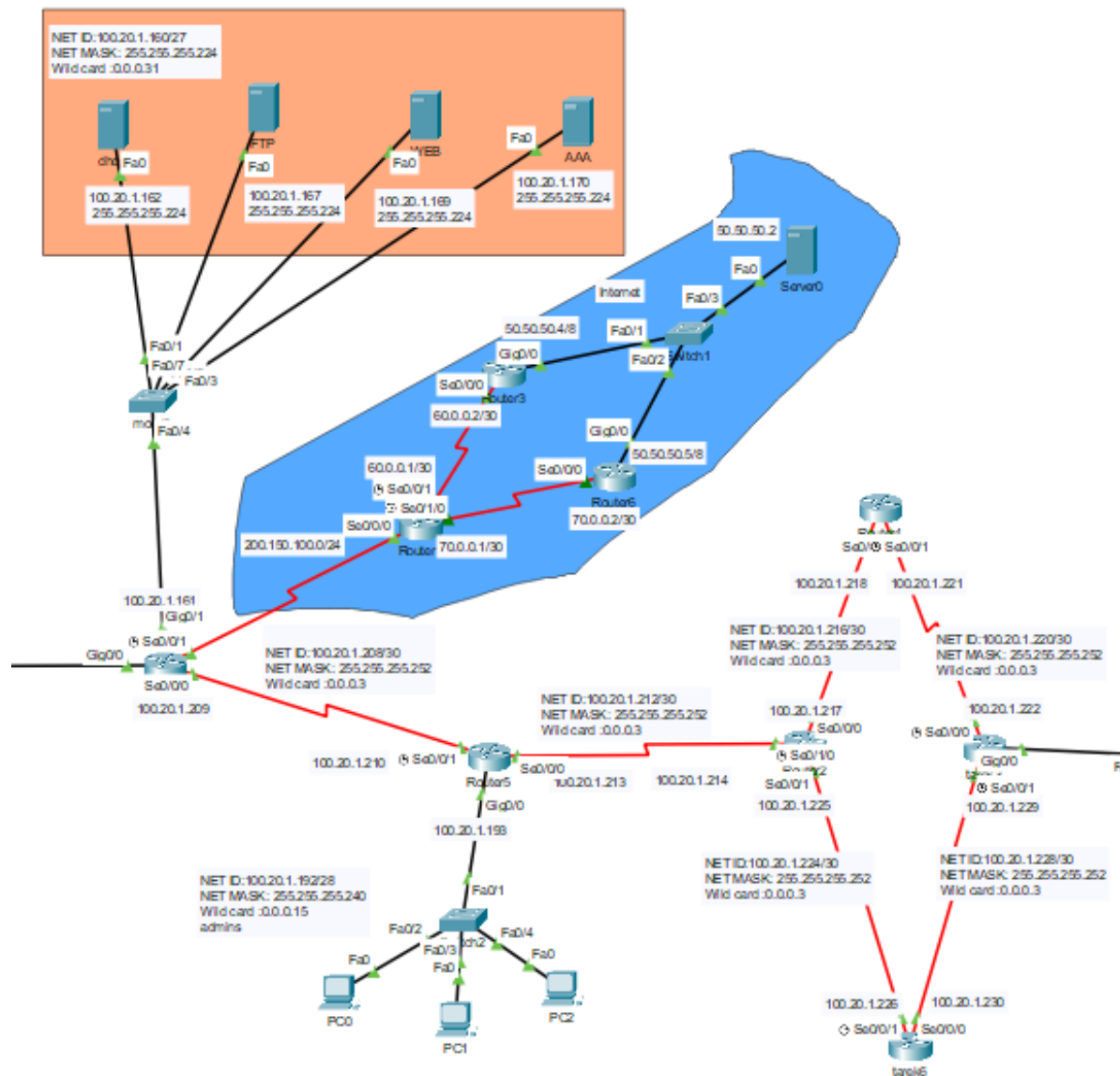
First branch:



second branch:



Network between branches and internet:



Variable length subnet masking (VLSM)

is a network addressing and design methodology that permits the division of an IP address space into subnets of different sizes.

Unlike fixed length subnetting, VLSM allocates address blocks to subnets based on their specific host requirements, using different subnet masks for different subnets within the same classful network.

We had the main network 100.20.0.0/6 then we divide it to 18 subnets

	Network Id	Subnet mask	Range of IPs	No. of hosts
Subnet 1	100.20.0.0/26	255.255.255.192	100.20.0.1 100.20.0.62	62
Subnet 2	100.20.0.64/26	255.255.255.192	100.20.0.65 100.20.0.126	62
Subnet 3	100.20.0.128/26	255.255.255.192	100.20.0.129 100.20.0.191	62
Subnet 4	100.20.0.192/26	255.255.255.192	100.20.0.193 100.20.0.254	62
Subnet 5	100.20.1.0/26	255.255.255.192	100.20.1.1 100.20.1.62	62
Subnet 6	100.20.1.64/27	255.255.255.224	100.20.1.65 100.20.1.94	30
Subnet 7	100.20.1.96/27	255.255.255.224	100.20.1.97 100.20.1.126	30
Subnet 8	100.20.1.128/27	255.255.255.224	100.20.1.129 100.20.1.158	30
Subnet 9	100.20.1.160/27	255.255.255.224	100.20.1.161 100.20.1.190	30

Subnet 10	100.20.1.192/28	255.255.255.240	100.20.1.193 100.20.1.206	14
Subnet 11	100.20.1.208/30	255.255.255.252	100.20.1.209 100.20.1.210	2
Subnet 12	100.20.1.212/30	255.255.255.252	100.20.1.213 100.20.1.214	2
Subnet 13	100.20.1.216/30	255.255.255.252	100.20.1.217 100.20.1.218	2
Subnet 14	100.20.1.220/30	255.255.255.252	100.20.1.221 100.20.1.222	2
Subnet 15	100.20.1.224/30	255.255.255.252	100.20.1.225 100.20.1.226	2
Subnet 16	100.20.1.228/30	255.255.255.252	100.20.1.229 100.20.1.230	2

VLANs

Virtual Local Area Networks (VLANs) allow network administrators to logically divide a single physical network into multiple isolated segments. This separation improves security, reduces broadcast traffic, and enhances network performance without the need for additional hardware. Devices inside the same VLAN can communicate directly, while communication between VLANs requires a Layer 3 device, such as a router or Layer 3 switch. VLANs are widely used in organizations to organize users, protect sensitive data, and simplify network management.

VLAN 10	100.20.0.0/26	255.255.255.192
VLAN 20	100.20.0.64/26	255.255.255.192
VLAN 30	100.20.0.128/26	255.255.255.192
VLAN 40	100.20.0.192/26	255.255.255.192
VLAN 50	100.20.1.0/26	255.255.255.192
VLAN 60	100.20.1.64/27	255.255.255.224
VLAN 70	100.20.1.96/27	255.255.255.224
VLAN 80	100.20.1.128/27	255.255.255.224

VLANs of first branch:

VLAN	Name	Status	Ports
1	default	active	Po2, Fa0/8, Fa0/9, Fa0/10 Fa0/11, Fa0/12, Fa0/13, Fa0/14 Fa0/15, Fa0/16, Fa0/17, Fa0/18 Fa0/19, Fa0/20, Fa0/21, Fa0/22 Fa0/23, Fa0/24, Gig0/1, Gig0/2
10	VLAN0010	active	
20	VLAN0020	active	
30	VLAN0030	active	
40	VLAN0040	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

VLANs of first branch:

morsi4#show vlan

VLAN	Name	Status	Ports
1	default	active	Fa0/8, Fa0/9, Fa0/10, Fa0/11 Fa0/12, Fa0/13, Fa0/14, Fa0/15 Fa0/16, Fa0/17, Fa0/18, Fa0/19 Fa0/20, Fa0/21, Fa0/22, Fa0/23 Fa0/24, Gig0/1, Gig0/2
50	VLAN0050	active	
60	VLAN0060	active	
70	VLAN0070	active	
80	VLAN0080	active	
1002	fddi-default	active	
1003	token-ring-default	active	
1004	fddinet-default	active	
1005	trnet-default	active	

Etherchannel

is a port-link aggregation technology, primarily associated with Cisco switches, that bundles multiple physical Ethernet links (between two to eight active ports) into a single logical link, often called a Port Channel. This process provides two major benefits: significantly increased bandwidth and fault tolerance (redundancy). Since the individual links are seen by the network as one logical connection, the Spanning Tree Protocol (STP) only sees a single link, which prevents it from blocking redundant physical connections and allows all links in the bundle to forward traffic simultaneously, distributing the load across them.

How we made it

we made in both branches

```
!
interface FastEthernet0/1
  switchport mode trunk
  channel-group 1 mode active
!
interface FastEthernet0/2
  switchport mode trunk
  channel-group 1 mode active
!
interface FastEthernet0/3
  switchport mode trunk
  channel-group 2 mode active
!
interface FastEthernet0/4
  switchport mode trunk
  channel-group 2 mode active
!
interface FastEthernet0/5
  switchport mode trunk
  channel-group 3 mode active
!
interface FastEthernet0/6
  switchport mode trunk
  channel-group 3 mode active
,
```

```
!
interface Port-channel1
  switchport mode trunk
!
interface Port-channel2
  switchport mode trunk
!
interface Port-channel3
  switchport mode trunk
!
```

we made interfaces trunk and activated Etherchannel on them we made three groups on for each switch

As the Access Layer switches are connected to the Distribution Layer switches using EtherChannel (link aggregation). This critical implementation provides high-speed aggregation and immediate link redundancy, ensuring that if any single physical cable connecting these two layers fails, traffic continues without interruption, thus maintaining the robust uptime required for our hospital operations

Inter VLAN routing

Inter-VLAN routing enables communication between devices in different VLANs. Since VLANs are isolated by default, a Layer 3 device like a router or Layer 3 switch is required to route traffic between them. This allows networks to stay segmented for security while still allowing necessary connectivity between departments and services

How we made it

```
interface GigabitEthernet0/0.10
  encapsulation dot1Q 10
  ip address 100.20.0.1 255.255.255.192
  ip helper-address 100.20.1.162
  ip nat inside
!
interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 100.20.0.65 255.255.255.192
  ip helper-address 100.20.1.162
  ip nat inside
!
interface GigabitEthernet0/0.30
  encapsulation dot1Q 30
  ip address 100.20.0.129 255.255.255.192
  ip helper-address 100.20.1.162
  ip nat inside
!
```

```

!
interface GigabitEthernet0/0.40
 encapsulation dot1Q 40
 ip address 100.20.0.193 255.255.255.192
 ip helper-address 100.20.1.162
 ip nat inside
!
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 100.20.1.1 255.255.255.192
 ip helper-address 100.20.1.162
!
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 100.20.1.65 255.255.255.224
 ip helper-address 100.20.1.162
!
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 100.20.1.97 255.255.255.224
 ip helper-address 100.20.1.162
!
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 100.20.1.129 255.255.255.224
 ip helper-address 100.20.1.162

```

Then after we made inter vlan routing we verify it by going inside a pc in vlan and ping on another vlan and the ping succeeded

As we used ROAS

Router-on-a-Stick (ROAS) is an inexpensive Inter-VLAN Routing method. It uses one router interface, logically split into subinterfaces (one per VLAN), connected to a switch via a single 802.1Q trunk link. This link is a bottleneck as all inter-VLAN traffic must pass through it


```
C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address.....: FE80::209:7CFF:FE7C:3B9E
    IPv6 Address.....: ::
    IPv4 Address.....: 100.20.0.70
    Subnet Mask.....: 255.255.255.192
    Default Gateway.....: ::
                           100.20.0.65
```

```
C:\>ping 100.20.0.8

Pinging 100.20.0.8 with 32 bytes of data:

Reply from 100.20.0.8: bytes=32 time=1ms TTL=127
Reply from 100.20.0.8: bytes=32 time<1ms TTL=127
Reply from 100.20.0.8: bytes=32 time<1ms TTL=127
Reply from 100.20.0.8: bytes=32 time<1ms TTL=127

Ping statistics for 100.20.0.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

This example is for ping on a pc in VLAN 10 from a pc in VLAN 20

Open short path first (OSPF)

OSPF is a dynamic routing protocol used to share routing information between routers inside the same network (internal gateway protocol). It uses link-state advertisements to build a complete map of the network and chooses the best path based on cost (bandwidth). OSPF quickly adapts to network changes, supports large and complex topologies, and divides networks into areas to optimize performance and reduce overhead

How we made it

```
router ospf 1
 log-adjacency-changes
 network 100.20.0.0 0.0.0.63 area 0
 network 100.20.0.64 0.0.0.63 area 0
 network 100.20.0.128 0.0.0.63 area 0
 network 100.20.0.192 0.0.0.63 area 0
 network 100.20.1.160 0.0.0.31 area 0
 network 100.20.1.208 0.0.0.3 area 0
 default-information originate
!
```

This is how it's done on first router connected to first branch


```

Gateway of last resort is 0.0.0.0 to network 0.0.0.0

  100.0.0.0/8 is variably subnetted, 22 subnets, 5 masks
C    100.20.0.0/26 is directly connected, GigabitEthernet0/0.10
L    100.20.0.1/32 is directly connected, GigabitEthernet0/0.10
C    100.20.0.64/26 is directly connected, GigabitEthernet0/0.20
L    100.20.0.65/32 is directly connected, GigabitEthernet0/0.20
C    100.20.0.128/26 is directly connected, GigabitEthernet0/0.30
L    100.20.0.129/32 is directly connected, GigabitEthernet0/0.30
C    100.20.0.192/26 is directly connected, GigabitEthernet0/0.40
L    100.20.0.193/32 is directly connected, GigabitEthernet0/0.40
O    100.20.1.0/26 [110/257] via 100.20.1.210, 00:23:41, Serial0/0/0
O    100.20.1.64/27 [110/257] via 100.20.1.210, 00:23:41, Serial0/0/0
O    100.20.1.96/27 [110/257] via 100.20.1.210, 00:23:41, Serial0/0/0
O    100.20.1.128/27 [110/257] via 100.20.1.210, 00:23:41, Serial0/0/0
C    100.20.1.160/27 is directly connected, GigabitEthernet0/1
L    100.20.1.161/32 is directly connected, GigabitEthernet0/1
O    100.20.1.192/28 [110/65] via 100.20.1.210, 00:23:51, Serial0/0/0
C    100.20.1.208/30 is directly connected, Serial0/0/0
L    100.20.1.209/32 is directly connected, Serial0/0/0
O    100.20.1.212/30 [110/128] via 100.20.1.210, 00:23:51, Serial0/0/0
O    100.20.1.216/30 [110/192] via 100.20.1.210, 00:23:51, Serial0/0/0
O    100.20.1.220/30 [110/256] via 100.20.1.210, 00:23:41, Serial0/0/0
O    100.20.1.224/30 [110/192] via 100.20.1.210, 00:23:51, Serial0/0/0
O    100.20.1.228/30 [110/256] via 100.20.1.210, 00:23:41, Serial0/0/0
  200.150.100.0/24 is variably subnetted, 2 subnets, 2 masks
C    200.150.100.0/24 is directly connected, Serial0/0/1
L    200.150.100.1/32 is directly connected, Serial0/0/1
S*   0.0.0.0/0 is directly connected, Serial0/0/1

tarek0#

```

This is the routing table of the router connected to the first branch and we can see that it sees all the networks in our design then OSPF done it's job

```

Gateway of last resort is 100.20.1.221 to network 0.0.0.0

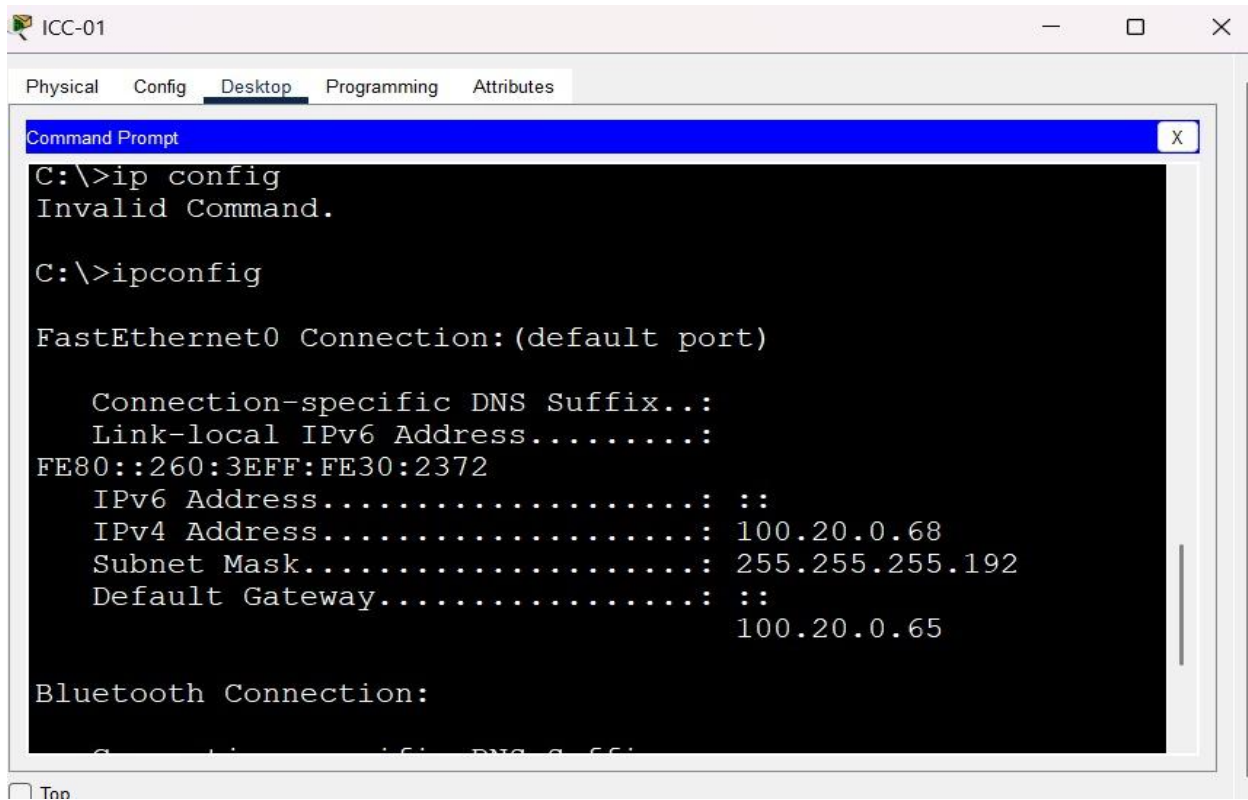
  100.0.0.0/8 is variably subnetted, 22 subnets, 5 masks
O    100.20.0.0/26 [110/257] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/257] via 100.20.1.230, 00:27:07, Serial0/0/1
O    100.20.0.64/26 [110/257] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/257] via 100.20.1.230, 00:27:07, Serial0/0/1
O    100.20.0.128/26 [110/257] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/257] via 100.20.1.230, 00:27:07, Serial0/0/1
O    100.20.0.192/26 [110/257] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/257] via 100.20.1.230, 00:27:07, Serial0/0/1
C    100.20.1.0/26 is directly connected, GigabitEthernet0/0.50
L    100.20.1.1/32 is directly connected, GigabitEthernet0/0.50
C    100.20.1.64/27 is directly connected, GigabitEthernet0/0.60
L    100.20.1.65/32 is directly connected, GigabitEthernet0/0.60
C    100.20.1.96/27 is directly connected, GigabitEthernet0/0.70
L    100.20.1.97/32 is directly connected, GigabitEthernet0/0.70
C    100.20.1.128/27 is directly connected, GigabitEthernet0/0.80
L    100.20.1.129/32 is directly connected, GigabitEthernet0/0.80
O    100.20.1.160/27 [110/257] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/257] via 100.20.1.230, 00:27:07, Serial0/0/1
O    100.20.1.192/28 [110/193] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/193] via 100.20.1.230, 00:27:07, Serial0/0/1
O    100.20.1.208/30 [110/256] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/256] via 100.20.1.230, 00:27:07, Serial0/0/1
O    100.20.1.212/30 [110/192] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/192] via 100.20.1.230, 00:27:07, Serial0/0/1
O    100.20.1.216/30 [110/128] via 100.20.1.221, 00:27:17, Serial0/0/0
C    100.20.1.220/30 is directly connected, Serial0/0/0
L    100.20.1.222/32 is directly connected, Serial0/0/0
O    100.20.1.224/30 [110/128] via 100.20.1.230, 00:27:17, Serial0/0/1
C    100.20.1.228/30 is directly connected, Serial0/0/1
L    100.20.1.229/32 is directly connected, Serial0/0/1
O*E2 0.0.0.0/0 [110/1] via 100.20.1.221, 00:27:07, Serial0/0/0
    [110/1] via 100.20.1.230, 00:27:07, Serial0/0/1

```

tarek3#

This is the routing table of the router connected to second branch and as we can see it shows all the networks in our design also

Now we ping from pc on VLAN in first branch to VLAN in second branch and it succeeds



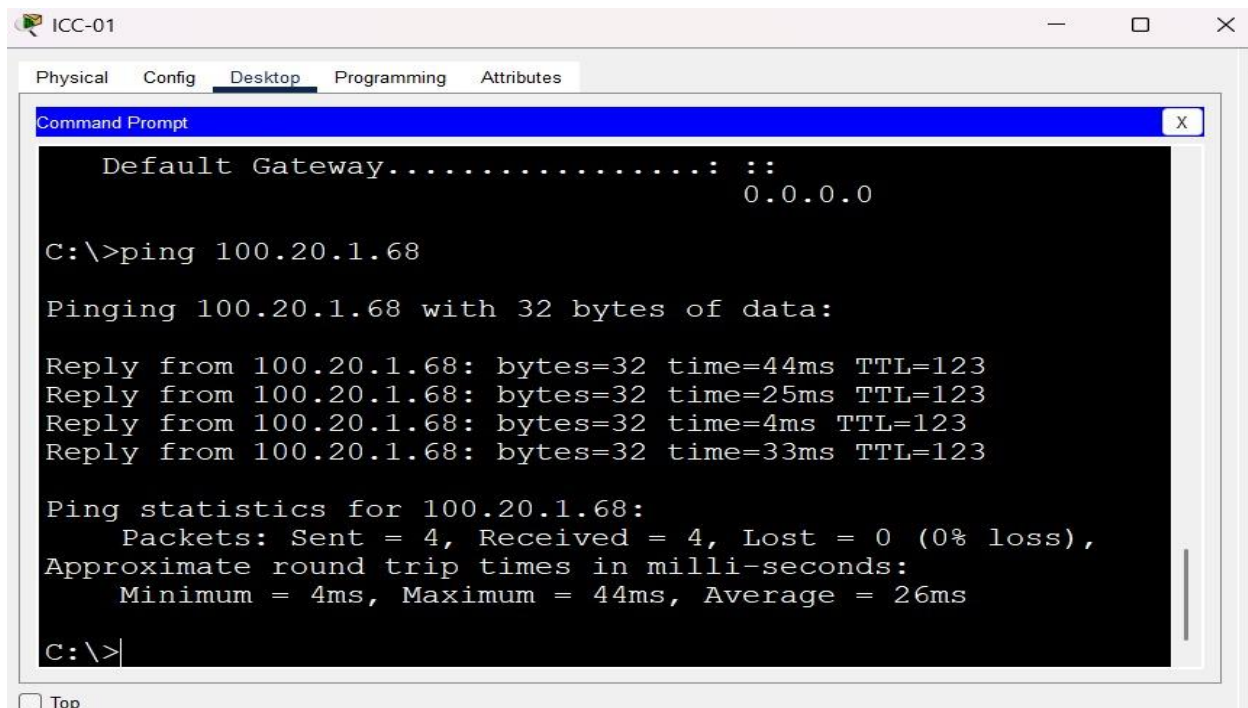
```
ICC-01
Physical Config Desktop Programming Attributes
Command Prompt
C:\>ip config
Invalid Command.

C:\>ipconfig

FastEthernet0 Connection:(default port)

    Connection-specific DNS Suffix...:
    Link-local IPv6 Address...: FE80::260:3EFF:FE30:2372
    IPv6 Address...: ::
    IPv4 Address...: 100.20.0.68
    Subnet Mask...: 255.255.255.192
    Default Gateway...: 100.20.0.65

Bluetooth Connection:
```



```
ICC-01
Physical Config Desktop Programming Attributes
Command Prompt

    Default Gateway...: 0.0.0.0

C:\>ping 100.20.1.68

Pinging 100.20.1.68 with 32 bytes of data:

Reply from 100.20.1.68: bytes=32 time=44ms TTL=123
Reply from 100.20.1.68: bytes=32 time=25ms TTL=123
Reply from 100.20.1.68: bytes=32 time=4ms TTL=123
Reply from 100.20.1.68: bytes=32 time=33ms TTL=123

Ping statistics for 100.20.1.68:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 4ms, Maximum = 44ms, Average = 26ms

C:\>
```

Dynamic Host Configuration Protocol DHCP

DHCP is a network service that automatically assigns IP addresses and other network settings (like subnet mask, gateway, and DNS) to devices on a network. Instead of manually configuring each device, DHCP simplifies the process by dynamically providing the required information when a device connects. This makes network management easier, prevents IP conflicts, and ensures smooth, efficient device connectivity

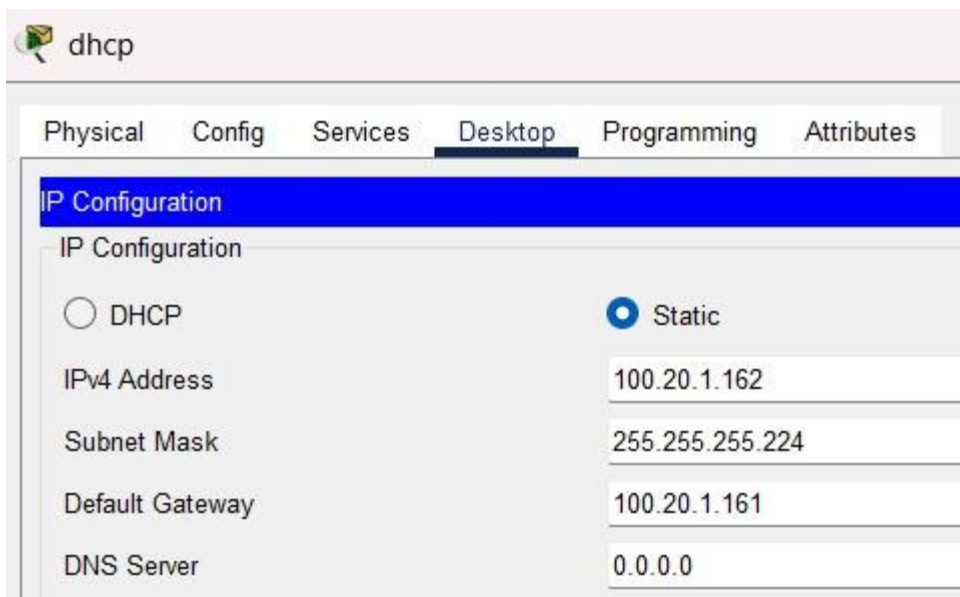
How we made it

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
admins	100.20.1.193	0.0.0.0	100.20.1.194	255.255.255.224	14	0.0.0.0	0.0.0.0
VLAN 80	100.20.1.129	0.0.0.0	100.20.1.130	255.255.255.224	30	0.0.0.0	0.0.0.0
VLAN 70	100.20.1.97	0.0.0.0	100.20.1.98	255.255.255.224	30	0.0.0.0	0.0.0.0
VLAN 60	100.20.1.65	0.0.0.0	100.20.1.66	255.255.255.224	30	0.0.0.0	0.0.0.0
VLAN 50	100.20.1.1	0.0.0.0	100.20.1.2	255.255.255.224	62	0.0.0.0	0.0.0.0
vlan 40	100.20.0.193	0.0.0.0	100.20.0.194	255.255.255.224	62	0.0.0.0	0.0.0.0
vlan 30	100.20.0.129	0.0.0.0	100.20.0.130	255.255.255.224	62	0.0.0.0	0.0.0.0
vlan 20	100.20.0.65	0.0.0.0	100.20.0.66	255.255.255.224	62	0.0.0.0	0.0.0.0
vlan 10	100.20.0.1	0.0.0.0	100.20.0.2	255.255.255.224	62	0.0.0.0	0.0.0.0

At first we made a pool for each VLAN and any thing has a pc to take their IPs from the DHCP server

And then we put IP helper address in all interfaces and sub interfaces of the routers which is the IP of the DHCP server to make all PCs able to take IPs from it

```
interface GigabitEthernet0/0.50
 encapsulation dot1Q 50
 ip address 100.20.1.1 255.255.255.192
 ip helper-address 100.20.1.162
!
interface GigabitEthernet0/0.60
 encapsulation dot1Q 60
 ip address 100.20.1.65 255.255.255.224
 ip helper-address 100.20.1.162
!
interface GigabitEthernet0/0.70
 encapsulation dot1Q 70
 ip address 100.20.1.97 255.255.255.224
 ip helper-address 100.20.1.162
!
interface GigabitEthernet0/0.80
 encapsulation dot1Q 80
 ip address 100.20.1.129 255.255.255.224
 ip helper-address 100.20.1.162
```



Where the IP we put on DHCP is 100.20.1.162

ACC-01

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☒ DHCP ☐ Static

IPv4 Address 100.20.0.194

Subnet Mask 255.255.255.192

Default Gateway 100.20.0.193

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:16FF:FE4D:34CA

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

As we can see a PC takes its IP from DHCP server

Authentication, Authorization, and Accounting AAA

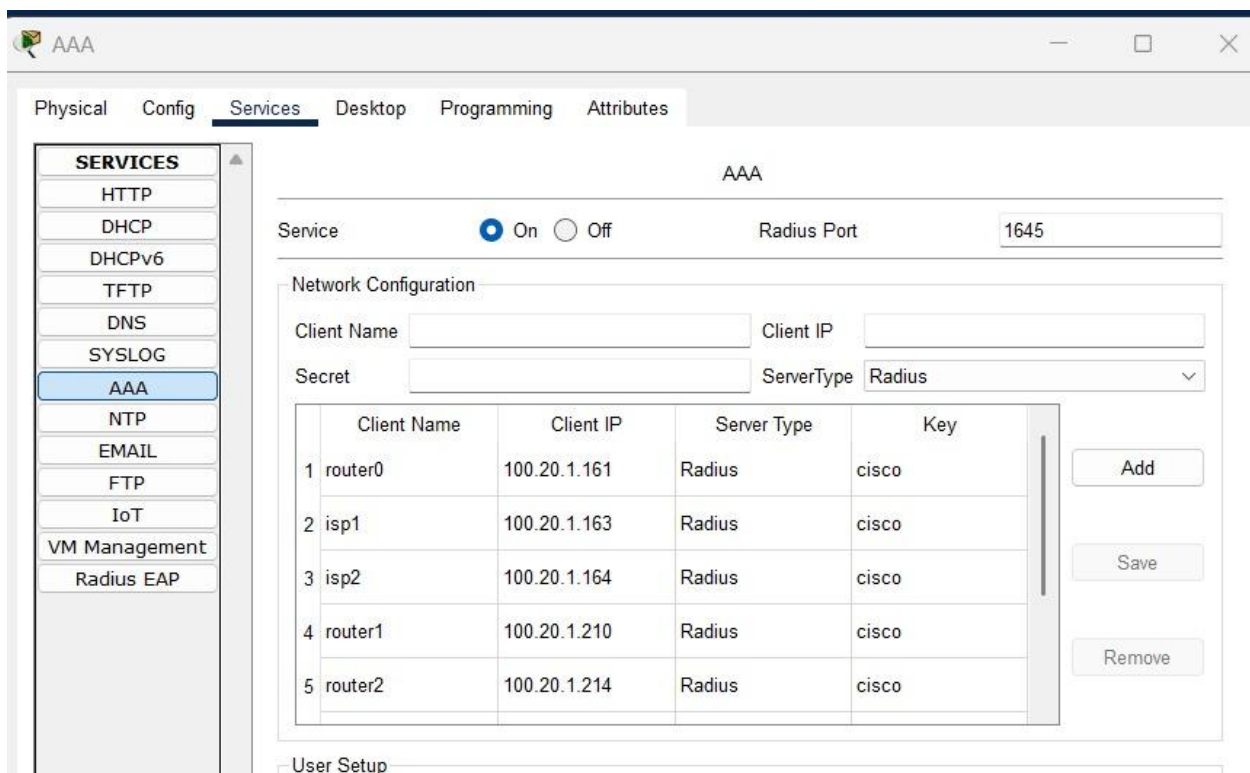
is a security framework used in networking to control and track user access to network resources.

Authentication verifies the identity of the user or device (e.g., username & password).

Authorization determines what the authenticated user is allowed to do (e.g., access specific devices or commands).

Accounting tracks user activities on the network for auditing and reporting.

How we made it



The screenshot shows a web-based configuration interface for AAA services. The left sidebar lists various services, with 'AAA' selected. The main panel displays the 'AAA' configuration page. At the top, there are tabs for 'Physical', 'Config', 'Services', 'Desktop', 'Programming', and 'Attributes'. The 'Services' tab is active. Below the tabs, there is a 'Service' section with a radio button set to 'On' and a 'Radius Port' field set to '1645'. The 'Network Configuration' section contains a table with columns for 'Client Name', 'Client IP', 'Server Type', and 'Key'. The table lists five entries: 1. router0 (100.20.1.161), 2. isp1 (100.20.1.163), 3. isp2 (100.20.1.164), 4. router1 (100.20.1.210), and 5. router2 (100.20.1.214). All entries have 'Radius' as the Server Type and 'cisco' as the Key. To the right of the table are buttons for 'Add', 'Save', and 'Remove'. At the bottom of the page, there is a 'User Setup' section.

	Client Name	Client IP	Server Type	Key
1	router0	100.20.1.161	Radius	cisco
2	isp1	100.20.1.163	Radius	cisco
3	isp2	100.20.1.164	Radius	cisco
4	router1	100.20.1.210	Radius	cisco
5	router2	100.20.1.214	Radius	cisco

At first we add all routers in our server with Radius type and choose cisco as key for all

User Setup

Username

Password

	Username	Password	
1	mohamed	1234	Add
2	tarek	1234	
			Save
			Remove

Then we add users that can access the routers and we put their passwords

```
aaa new-model
radius-server host 100.20.1.170 key cisco
aaa authentication login default group radius local
line console 0
login authentication default
exit
```

Then we put this configuration on all routers to make authentication occur

```
User Access Verification
```

```
Username: tarek
Password:
tarek5>
```

And that is the result of applying the configuration

File Transfer Protocol (FTP)

is a standard network protocol used to transfer computer files between a client and a server on a computer network. FTP is built on client-server model architecture and uses separate control and data connections between the client and the server. The control connection, typically over port 21, is used for transmitting commands (like logging in, changing directories, or listing files), while the data connection (often over port 20 for active mode) is used for the actual transfer of file data.

How we made it

The screenshot shows the 'Services' configuration page for FTP. The interface includes a sidebar with a list of services, a main configuration area for the selected service, and a file list at the bottom.

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP**
- IoT
- VM Management
- Radius EAP

FTP Configuration

Service: ☒ On ☐ Off

User Setup

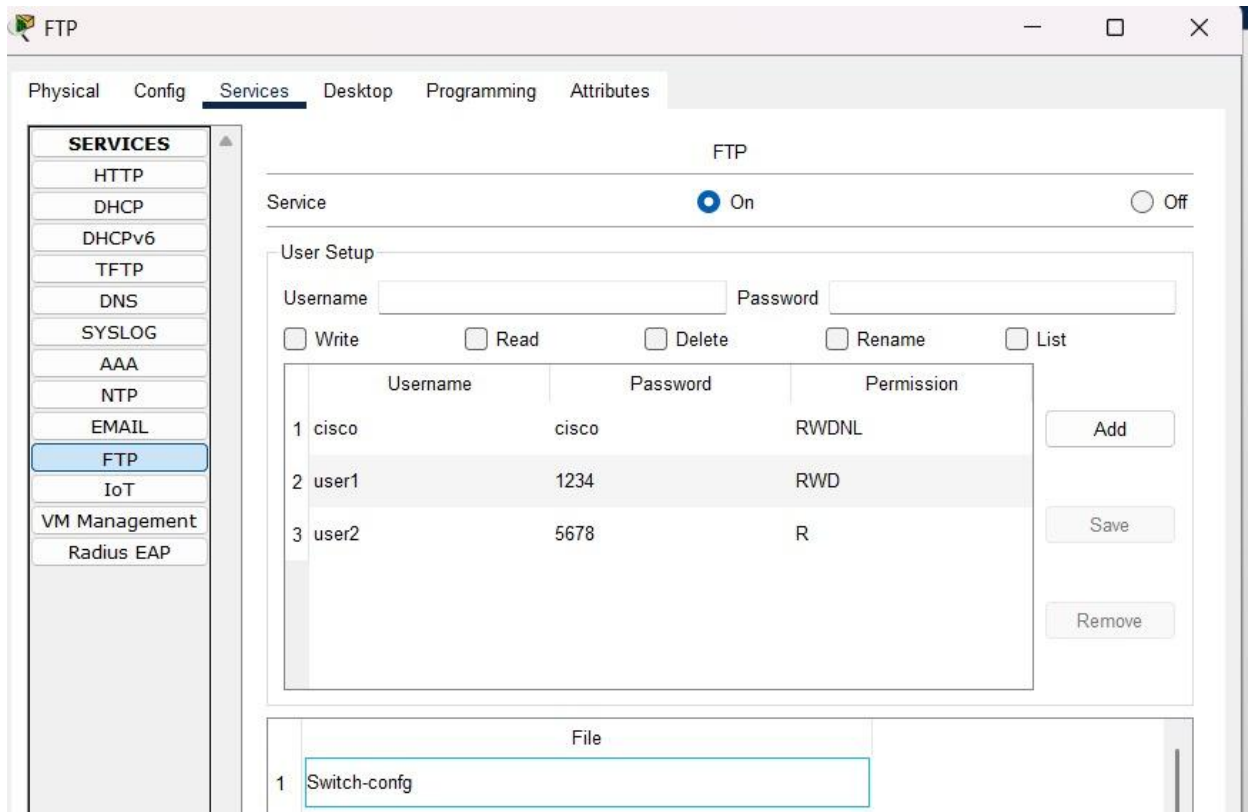
Username: Password:

☐ Write ☐ Read ☐ Delete ☐ Rename ☐ List

	Username	Password	Permission
1	cisco	cisco	RWDNL
2	user1	1234	RWD
3	user2	5678	R

File

28	ir800-universalk9-mz.SPA.155-3.M
29	ir800-universalk9-mz.SPA.156-3.M
30	ir800_yocto-1.7.2.tar
31	ir800_yocto-1.7.2_python-2.7.3.tar
32	pt1000-i-mz.122-28.bin
33	pt3000-i6q4l2-mz.121-22.EA4.bin
34	tarek0-config



We put user names and passwords on it and gave them permissions and we put configuration of one router and one switch (file 34 and file 1) on it put an IP on it which is 100.20.1.167

```
tarek0#copy startup-config ftp:
Address or name of remote host []? 100.20.1.167
Destination filename [tarek0-config]?

Writing startup-config...
[OK - 2399 bytes]

2399 bytes copied in 0.075 secs (31000 bytes/sec)
```

```
Switch#copy startup-config ftp:  
Address or name of remote host []? 100.20.1.167  
Destination filename [Switch-config]?  
  
Writing startup-config...  
[OK - 1176 bytes]  
  
1176 bytes copied in 0.015 secs (78000 bytes/sec)
```

As we copied files of router and switch to the FTP server

```
C:\>ftp 100.20.1.167  
Trying to connect...100.20.1.167  
Connected to 100.20.1.167  
220- Welcome to PT Ftp server  
Username:cisco  
331- Username ok, need password  
Password:  
230- Logged in  
(passive mode On)  
ftp>ls
```

Then we entered it from pc

IP Static Network Address Translation (Static NAT)

is a type of address translation that creates a fixed, one-to-one mapping between a single internal private IP address and a single external public IP address. Unlike other forms of NAT that allow multiple internal devices to share one public IP, Static NAT reserves a unique public address for a specific internal device permanently. This fixed mapping is configured manually on a router or firewall and is immediately stored in the NAT translation table, ensuring that the internal device is always represented by the same public IP address when communicating externally.

How we made it

```
!
ip nat inside source list 2 interface Serial0/0/1 overload
ip nat inside source static 100.20.1.169 200.150.100.1
ip classless
ip route 0.0.0.0 0.0.0.0 Serial0/0/1
!
ip flow-export version 9
!
!
access-list 2 permit any
!
```

We made access list to permit PCs to reach the public ip (50.50.50.2) through static NAT then we choose the interface serial 0/0/1 to be the way for the public ip which is the internet in our design and for another command we choose the inside global ip in the outside network (internet)

```

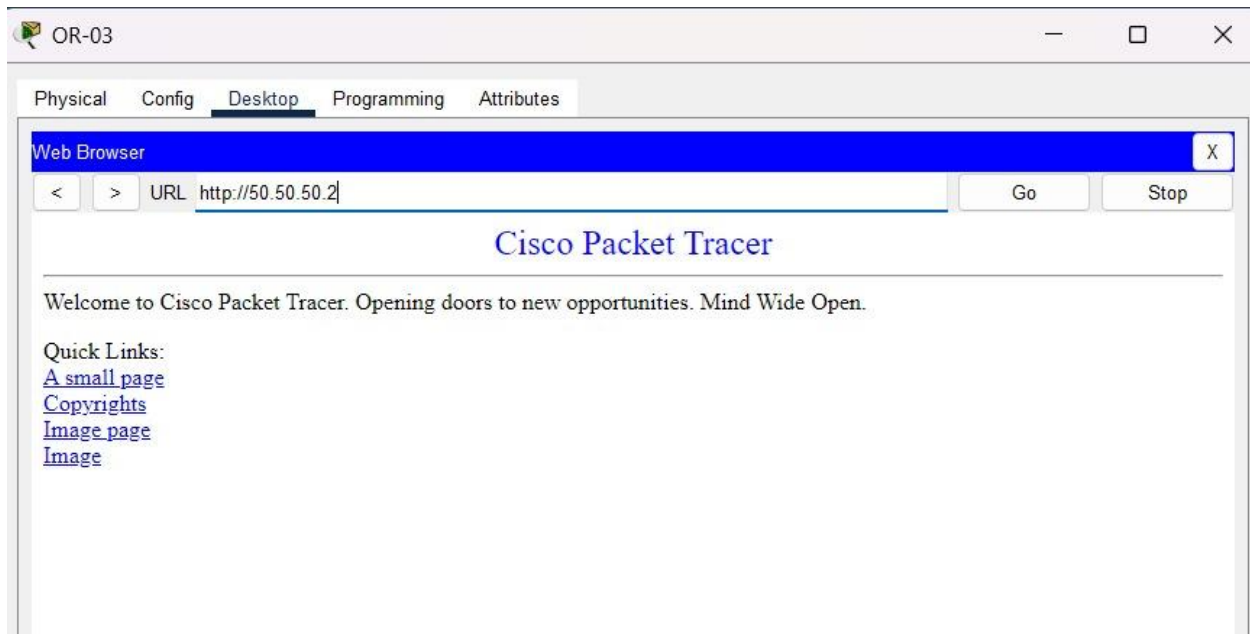
!
interface Serial0/0/0
 ip address 100.20.1.209 255.255.255.252
 ip nat inside
!
interface Serial0/0/1
 ip address 200.150.100.1 255.255.255.0
 ip nat outside
 clock rate 2000000
!
interface GigabitEthernet0/0
 no ip address
 ip nat inside
 duplex auto
 speed auto
!

```

then we choose inside and outside in interfaces of the router

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	AAA	Server0	ICMP		0.000	N	0	(edit)	
	Failed	Server0	AAA	ICMP		0.000	N	1	(edit)	

And as we can see when we ping from AAA server which is inside our network to server that has the ip of 50.50.50.2 it reaches it and when this server try to reach inside it fails which verify that static NAT is successful



and when we write the IP in our web browser it opens the page

The Hot Standby Router Protocol (HSRP)

is a Cisco proprietary First Hop Redundancy Protocol (FHRP) designed to prevent a single point of failure at the default gateway for hosts on a local area network (LAN). HSRP allows a group of routers to cooperate and present themselves as a single virtual router. This virtual router has a shared Virtual IP Address (VIP) and Virtual MAC Address, which is configured as the default gateway on all end devices. Within the group, one router is elected the Active Router and is responsible for forwarding all traffic destined for the VIP, while a second router is designated as the Standby Router, constantly monitoring the active device and ready to take over



How we made it

In our design we made it in the internet as we put two routers to reach the 50.50.50.2 server one active and the other stand by

The image displays three screenshots of network configuration interfaces. The first two are for routers in a CLI environment, and the third is for a web-server in a GUI environment.

Router3
Physical Config CLI Attributes
IOS Command Line Interface

```
!
interface GigabitEthernet0/0
 ip address 50.50.50.4 255.0.0.0
 duplex auto
 speed auto
 standby 1 ip 50.50.50.1
!
```

Router6
Physical Config CLI Attributes
IOS Command Line Int

```
!
interface GigabitEthernet0/0
 ip address 50.50.50.5 255.0.0.0
 duplex auto
 speed auto
 standby 1 ip 50.50.50.1
!
```

web-server
Physical Config Services Desktop Programming Attribute

IP Configuration	
IP Configuration	
☐ DHCP	☒ Static
IPv4 Address	50.50.50.2
Subnet Mask	255.0.0.0
Default Gateway	50.50.50.1
DNS Server	0.0.0.0

we configured HSRP configuration on both routers and assigned the same virtual IP to both of them and adjust the default gateway of the server to be this virtual IP as if one router failed the other takes responsibility

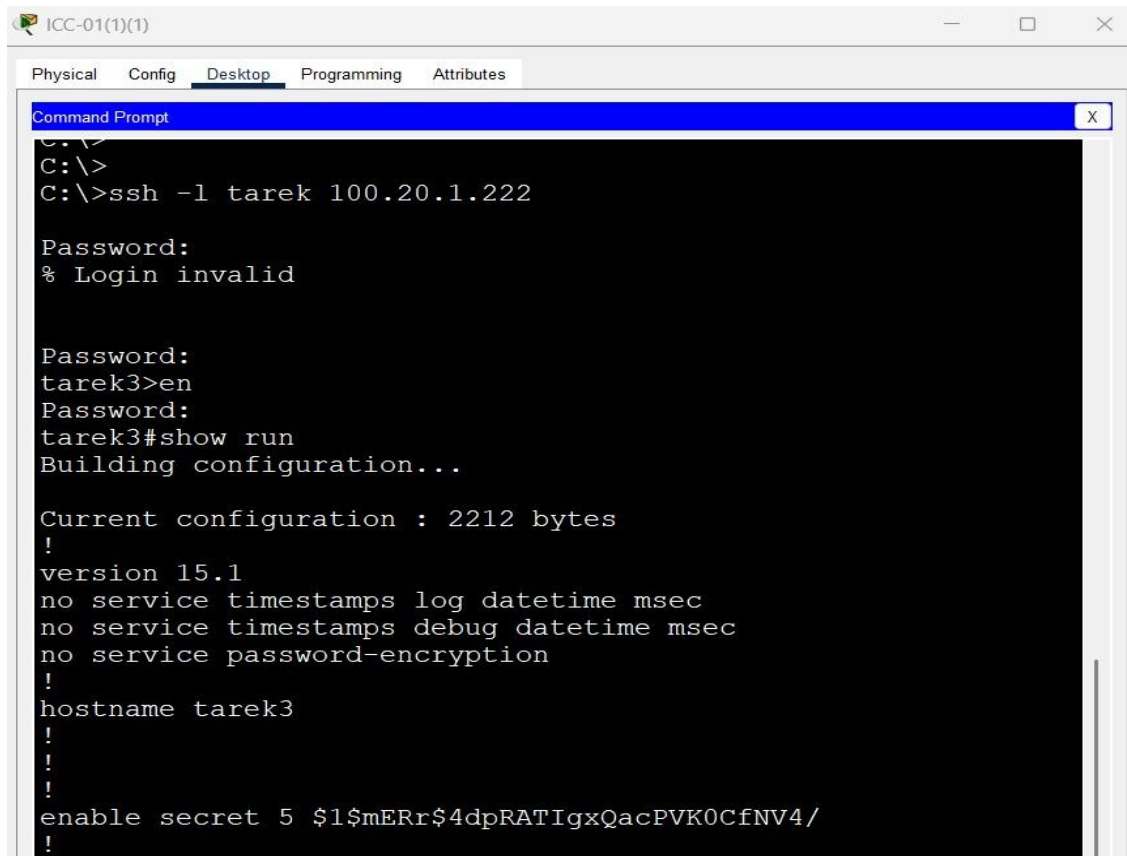
Secure Shell (SSH)

is a cryptographic network protocol used to operate network services securely over an unsecured network. Its most common application is for remote command-line login and execution, replacing insecure protocols like Telnet and rlogin. SSH uses a client-server model and employs strong encryption to ensure that all data transmitted, including passwords and commands, is protected from eavesdropping and hijacking. Beyond secure remote access, SSH is also widely used for secure file transfers via protocols like SFTP (SSH File Transfer Protocol) and SCP (Secure Copy Protocol), and for managing networking hardware and servers securely.

How we made it

```
!
line vty 0 4
  login authentication default
  transport input ssh
!
```

We put this configuration on the router connected to second branch



```
ICC-01(1)(1)
Physical Config Desktop Programming Attributes
Command Prompt
C:\>
C:\>ssh -l tarek 100.20.1.222

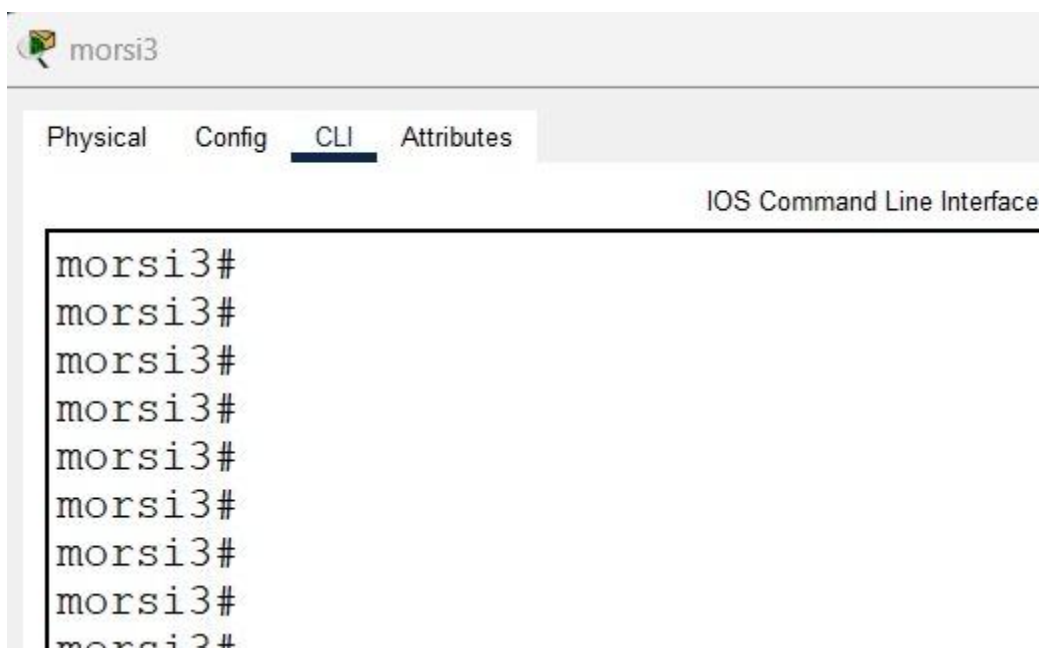
Password:
% Login invalid

Password:
tarek3>en
Password:
tarek3#show run
Building configuration...

Current configuration : 2212 bytes
!
version 15.1
no service timestamps log datetime msec
no service timestamps debug datetime msec
no service password-encryption
!
hostname tarek3
!
!
!
enable secret 5 $1$mERr$4dpRATigxQacPVKOCfNV4/
!
```

When we access the router from a pc through ssh it succussed

We put team member name on switches and the other on the routers



```
morsi3
Physical Config CLI Attributes
IOS Command Line Interface

morsi3#
morsi3#
morsi3#
morsi3#
morsi3#
morsi3#
morsi3#
morsi3#
morsi3#
```



Router5

Physical

Config

CLI

Attributes

IOS Command Line Interface

Password:

tarek5>

tarek5>

tarek5>

tarek5>

tarek5>

tarek5>

tarek5>

tarek5>